Week 3: Sentiment & News Alphas

- Dates: 2026-10-20 (midterm week)
- Scope: BRAIN Tutorial Chapter 5 — Sentiment & News Alphas (3/3 complete)
- Simulations: 4, 1 submitted (6 in total so far)
- Common settings: USA · D1 · TOP3000 · Subindustry · Truncation 0.08 · NaN handling On · Test period 0 (full in-sample period)

> Chapter 6 (reaching 10,000 Challenge points) is next week's scope. Work I started on it early will be written up in week 4.

Summary

Chapter 5 introduced two new kinds of data: **social-media sentiment** (how much a stock is talked about online) and **news** (whether going long or short paid off after each news release). The sentiment signal looked mediocre overall but hid a regime change: it worked strongly through 2021 and then flipped direction. News data is stored as *vectors* — several items per stock per day — so it has to be collapsed with `vec_avg` before regular operators apply. After that, a single conditional expression switching between momentum and reversal produced a stable alpha, which I submitted.

Roadmap

```
Step 1  social buzz vs volume (ts_regression)    Sharpe -1.06 → flipped 1.06, direction reverses
in 2022
Step 2  news "long-favoured" → momentum, else 5-day reversal
                                                   Sharpe 1.95, Fitness 1.12 (Decay 4)
Step 3  same with Decay 8                          Sharpe 1.91, Fitness 1.25  --> SUBMITTED
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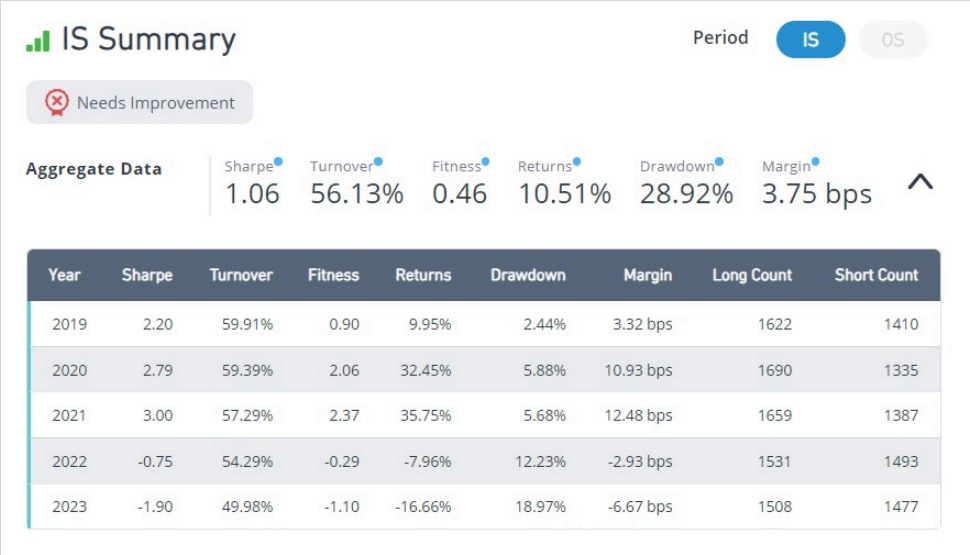
1. Sentiment: social buzz versus trading volume

Idea: if a stock is discussed online far more than its trading volume would suggest, attention may be overheated.

`ts_regression(y, x, d)` fits a line of `y` on `x` over the last `d` days and returns the residual — here, "how much more buzz than volume explains". `scl12_buzz` measures how much a stock is mentioned online.

Expression	Sharpe	2019	2020	2021	2022	2023
`ts_regression(scl12_buzz, volume, 250)`	-1.06	-2.20	-2.79	-3.00	+0.75	+1.90
`-ts_regression(scl12_buzz, volume, 250)`	1.06	**+2.20**	**+2.79**	**+3.00**	-0.75	-1.90

As the tutorial suggests, I flipped the sign when the direction was wrong. The flipped signal was very strong through 2021 (excess attention → price falls) and then **reversed from 2022**. The aggregate Sharpe of 1.06 hides this; only the yearly table shows it. It also failed the checks (56% turnover, one-day weight concentration, weak sub-universe), so I moved on.

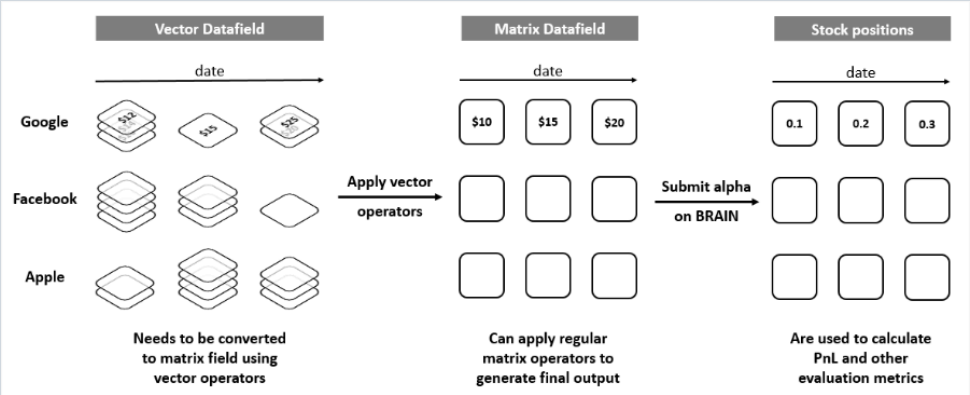


Year	Sharpe	Turnover	Fitness	Returns	Drawdown	Margin	Long Count	Short Count
2019	2.20	59.91%	0.90	9.95%	2.44%	3.32 bps	1622	1410
2020	2.79	59.39%	2.06	32.45%	5.88%	10.93 bps	1690	1335
2021	3.00	57.29%	2.37	35.75%	5.68%	12.48 bps	1659	1387
2022	-0.75	54.29%	-0.29	-7.96%	12.23%	-2.93 bps	1531	1493
2023	-1.90	49.98%	-1.10	-16.66%	18.97%	-6.67 bps	1508	1477

2. News: vector data and a conditional alpha

News data covers earnings reports, analyst calls, product approvals and legal events. US news mostly comes out in the pre-market (4:00–9:30) or after-hours (16:00–20:00) sessions.

Because one stock can have several news items per day, the fields are **vectors**: one cell holds several values. Regular operators expect one value per stock per day, so a `vec_` operator such as `vec_avg` must collapse each day's items first.



`nws12_afterhsz_sl` tells whether going long or short was more profitable after each news release.

Idea: stocks whose recent news favoured longs show momentum; the rest tend to revert. Following the tutorial's recipe:

1. average each day's news values with `vec_avg`
2. smooth with a 60-day `ts_mean` to reduce daily noise
3. split the universe in half with `rank(...) > 0.5`
4. top half → constant long (the condition already captures momentum); bottom half → 5-day price reversal, joined with a conditional `? :`

Version	Sharpe	Turnover	Fitness	Drawdown	Margin
Decay 4 (tutorial step 2)	1.95	22.73%	1.12	3.25%	6.58 bps
Decay 8 (step 3)	1.91	**16.08%**	**1.25**	3.02%	**8.60 bps**

Yearly Sharpe for the Decay 4 version: 2.74 / 4.19 / 1.72 / 0.22 / 1.92. Fitness was close to the 1.0 cutoff, so — as in week 1 — I raised Decay: turnover fell by a third with almost no loss of Sharpe.

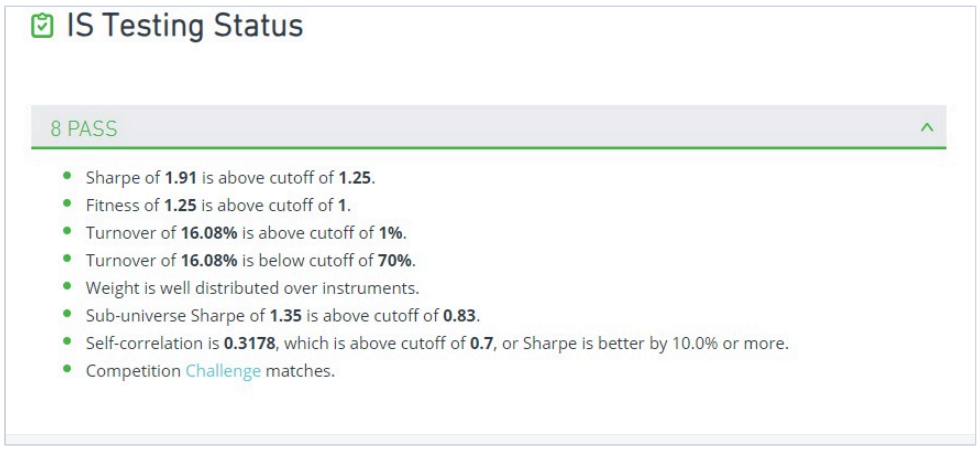
Submitted: news momentum / reversal ✓

Item	Detail
Chapter	Chapter 5, Step 3
Operators	`vec_avg`, `ts_mean`, `rank`, `ts_delta`, conditional `? :`
Data fields	`nws12_afterhsz_sl`, `close`
Settings	USA · TOP3000 · Delay 1 · Subindustry · Decay 8
Result (2019–2023)	**Sharpe 1.91 · Fitness 1.25 · Turnover 16.08%** · Returns 6.92% · Drawdown 3.02%
Checks	8 passed. Sub-universe Sharpe 1.35, self-correlation 0.32

> For submitted alphas I share the idea, operators and data fields, but not the exact expression.



Year	Sharpe	Turnover	Fitness	Returns	Drawdown	Margin	Long Count	Short Count
2019	2.70	16.24%	1.73	6.67%	2.02%	8.22 bps	1565	1450
2020	4.16	16.28%	4.09	15.72%	1.14%	19.32 bps	1577	1432
2021	1.59	16.00%	1.03	6.66%	2.48%	8.33 bps	1639	1423
2022	0.32	16.00%	0.09	1.39%	3.02%	1.73 bps	1634	1426
2023	1.66	15.90%	0.92	4.87%	1.17%	6.13 bps	1654	1421





What I learned this week

1. **Read the yearly table, not just the total.** The sentiment signal's Sharpe of 1.06 hid a sign flip in 2022.
2. **Flip a signal only when it is consistently wrong.** Flipping fixes a reversed direction, not a regime change.
3. **Shape the data first.** Vector data needs `vec_avg` before `ts_mean`, `rank` and the rest.
4. **One condition can combine two strategies.** News decides whether a stock gets the momentum leg or the reversal leg.
5. **When fitness is borderline, raise Decay.** 22.7% → 16.1% turnover lifted fitness from 1.12 to 1.25.

Next week

- Chapter 6: submit more alphas until the Challenge score reaches 10,000 (Gold)
- Look for new ingredients rather than variations of submitted alphas, since near-copies fail the self-correlation check

All experiments

#	Idea	Decay	Sharpe	Turnover	Fitness	Note
1	Buzz vs volume regression (250 days)	4	-1.06	56.13%	-0.46	wrong direction
2	#1 reversed	4	1.06	56.13%	0.46	flips in 2022
3	News-conditioned momentum / reversal	4	1.95	22.73%	1.12	8 checks passed
4	#3 with Decay 8	8	1.91	16.08%	1.25	submitted